# 3-3-26 (WEEK 2)

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## RECALL

The sample space $S$ is the set of all possible outcomes
An event is a subset $E \subseteq S$ of outcomes we are interested in

## UNIONS

Given events $E_1, \dots, E_n$, we say:

- $\bigcup_{i=1}^{\infty} E_i = \{x \in S \mid \exists\, i \text{ with } x \in E_i\}$
- $\bigcap_{i=1}^{\infty} E_i = \{x \in S \mid x \in E_i\ \forall i\}$
- $E_1, E_2, \dots$ are mutually exclusive (i.e. pairwise disjoint) if $E_i \cap E_j = \emptyset\ \forall i \neq j$

## COMPLEMENTS

Given $E \subseteq S$, the complement $E^c = \{x \in S \mid x \notin E\}$

## SET DIFFERENCES

Given $E, F \subseteq S$, $F \setminus E = \{x \in F \mid x \notin E\} = F \cap E^c$
The symmetric difference $E \triangle F = (E \setminus F) \cup (F \setminus E)$

## PROPOSITION (DE MORGAN'S LAWS)

Given events $E_1, E_2, \dots \subseteq S$, $\left(\bigcup_{i=1}^{\infty} E_i\right)^c = \bigcap_{i=1}^{\infty} (E_i^c)$ and $\left(\bigcap_{i=1}^{\infty} E_i\right)^c = \bigcup_{i=1}^{\infty} (E_i^c)$

Proof

① "$\left(\bigcup_{i=1}^{\infty} E_i\right)^c = \bigcap_{i=1}^{\infty} (E_i^c)$": By def, $\left(\bigcup_{i=1}^{\infty} E_i\right)^c = \{x \in S \mid x \notin \bigcup_{i=1}^{\infty} E_i\} = \{x \in S \mid \nexists\, i, \text{ s.t. } x \in E_i\} = \{x \in S \mid \forall i, x \notin E_i\} = \{x \in S \mid \forall i, x \in E_i^c\} = \bigcap_{i=1}^{\infty} (E_i^c)$ □

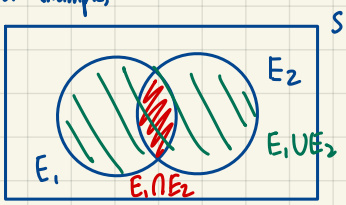
② "$\left(\bigcap_{i=1}^{\infty} E_i\right)^c = \bigcup_{i=1}^{\infty} (E_i^c)$": observe, for any $E$, $(E^c)^c = E$

$\therefore$ Let $F_i := E_i^c$, then $\left(\bigcup_{i=1}^{\infty} F_i\right)^c = \bigcap_{i=1}^{\infty} F_i^c = \bigcap_{i=1}^{\infty} (E_i^c)^c = \bigcap_{i=1}^{\infty} E_i$

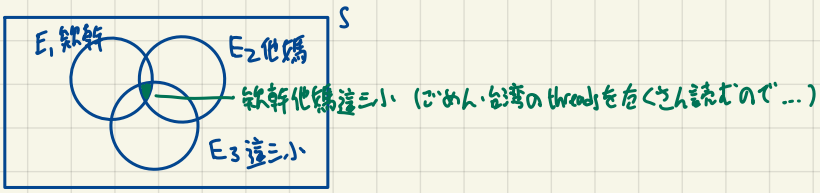
$\therefore \bigcap_{i=1}^{\infty} E_i = \left(\bigcup_{i=1}^{\infty} F_i\right)^c = \left(\bigcup_{i=1}^{\infty} (E_i^c)^c\right) \Rightarrow$ Take complement on both sides, $\left(\bigcap_{i=1}^{\infty} E_i\right)^c = \bigcup_{i=1}^{\infty} (E_i^c)$ □

## VENN DIAGRAMS

For example,




For 3 events,




## CAUTION

Make sure all intersections of events must be included in your diagram

## DEFINITION

Given a sample space $S$, a probability measure is a function $\mathbb{P} : \underbrace{2^S}_{\{E \mid E \subseteq S\}} \longrightarrow \mathbb{R}$ satisfying the following:

① For any event, $0 \leq \mathbb{P}(E) \leq 1$

② $\mathbb{P}(S) = 1$

③ [countable additivity] For any sequence of mutually exclusive events $E_1, E_2, E_3, \dots$, we must have $\mathbb{P}\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(E_i)$

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**DEFINITION**

Given a sample space $S$, and a probability measure $\mathbb{P}: 2^S \to \mathbb{R}$, we call $(S, \mathbb{P})$ a probability space

# PROPERTIES OF PROBABILITY SPACES

## PROPERTY 1

$\mathbb{P}(\emptyset) = 0$

Proof

Consider the following sequence of events: $E_1 = S$, $E_i = \emptyset \ \forall i \geq 2$

Observe that these are mutually exclusive: $\forall i \neq j$, $E_i \cap E_j = \emptyset$ since one of $E_i, E_j$ must be empty

By countable additivity, we know $\mathbb{P}(\bigcup_{i=1}^{\infty} E_i) = \sum_{i=1}^{\infty} \mathbb{P}(E_i)$

↳ $LHS = \mathbb{P}(\bigcup_{i=1}^{\infty} E_i) = \mathbb{P}(S) = 1$

↳ $RHS = \sum_{i=1}^{\infty} \mathbb{P}(E_i) = \mathbb{P}(S) + \sum_{i=2}^{\infty} \mathbb{P}(\emptyset) = 1 + \sum_{i=2}^{\infty} \mathbb{P}(\emptyset)$

$\therefore 1 = 1 + \sum_{i=2}^{\infty} \mathbb{P}(\emptyset) \Rightarrow \sum_{i=2}^{\infty} \mathbb{P}(\emptyset) = 0 \Rightarrow \mathbb{P}(\emptyset) = 0$ □

## PROPERTY 2 (FINITE UNIONS)

If $E_1, E_2, \ldots, E_n$ is a finite sequence of mutually exclusive events, then $\mathbb{P}(\bigcup_{i=1}^{n} E_i) = \sum_{i=1}^{n} \mathbb{P}(E_i)$

Proof

Consider the infinite sequence $E_1, E_2, \ldots, E_n, \emptyset, \emptyset, \ldots$, which is mutually exclusive

$\therefore$ By countable additivity, we have:

$$\begin{array}{ccc} \mathbb{P}(\bigcup_{i=1}^{\infty} E_i) & = & \sum_{i=1}^{\infty} \mathbb{P}(E_i) \\ \| & & \| \ [\mathbb{P}(\emptyset) = 0] \\ \mathbb{P}(\bigcup_{i=1}^{n} E_i) & = & \sum_{i=1}^{n} \mathbb{P}(E_i) \end{array}$$ □

## PROPERTY 3 (COMPLEMENTS)

For any event $E$, $\mathbb{P}(E^c) = 1 - \mathbb{P}(E)$

Proof

Consider the finite sequence: $E_1 = E$, $E_2 = E^c$. Note, $E \cap E^c = \emptyset$, $E \cup E^c = S$.

By finite additivity, $1 = \mathbb{P}(S) = \mathbb{P}(E \cup E^c) = \mathbb{P}(E) + \mathbb{P}(E^c)$

$\therefore \mathbb{P}(E^c) = 1 - \mathbb{P}(E)$ □

## PROPERTY 4 (MONOTONICITY)

If $E \subseteq F$ are events, then $P(E) \leq P(F)$

Proof

Notice, $F = E \sqcup (F \setminus E)$

By finite additivity, $\mathbb{P}(F) = \mathbb{P}(E) + \mathbb{P}(F \setminus E) \geq \mathbb{P}(E) + 0 = \mathbb{P}(E)$ □

## EXAMPLE

「何で不当に死んだよ!!」

We have an unfair die, which rolls a 1 with probability 0.1, 2 with probability 0.2, 3 with probability 0.25, 4 with probability 0.15, 5 with probability 0.17. This die has 6 sides labelled 1–6.

ⓐ What is the probability of rolling a 6?

ⓑ What is the probability of rolling a prime?

Solution

$S = \{1, 2, 3, 4, 5, 6\}$.

To define the probability over a countable sample space, it is enough to define the probability of individual outcomes $\{x\} \subseteq S$

↳ Because $\forall E \subseteq S$, $E = \bigcup_{x \in E} \{x\} \Rightarrow \mathbb{P}(E) = \sum_{x \in E} \mathbb{P}(\{x\})$

We have from our question: $\mathbb{P}(\{1\}) = 0.1$, $\mathbb{P}(\{2\}) = 0.2$, $\mathbb{P}(\{3\}) = 0.25$, $\mathbb{P}(\{4\}) = 0.15$, $\mathbb{P}(\{5\}) = 0.17$

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a) Notice, in our $S$, $\{1,2,3,4,5\}^c = \{6\}$

$\therefore \mathbb{P}(\{6\}) = 1 - \mathbb{P}(\{1,2,3,4,5\}) = 1 - \sum_{i=1}^{5} \mathbb{P}(\{i\}) = 1 - (0.1 + 0.2 + 0.25 + 0.15 + 0.17) = 0.13$

b) $\mathbb{P}(\{\text{rolling a prime}\}) = \mathbb{P}(\{2,3,5\}) = \mathbb{P}(\{2\} \cup \{3\} \cup \{5\}) = \mathbb{P}(\{2\}) + \mathbb{P}(\{3\}) + \mathbb{P}(\{5\}) = 0.62$

## EXAMPLE

Any uniform probability measure satisfies the axioms

Proof

Let $S$ be a finite nonempty sample place. We defined the uniform probability measure as $\mathbb{P}(E) := \frac{|E|}{|S|}$

① $\forall E \subseteq S$, $0 \leq |E| \leq |S| \Rightarrow 0 \leq \frac{|E|}{|S|} = \mathbb{P}(E) \leq 1$ ✓

② $\mathbb{P}(S) = \frac{|S|}{|S|} = 1$ ✓

③ If $E_1, E_2, \ldots$ are mutually exclusive

Goal: "$\mathbb{P}(\bigcup_{i=1}^{\infty} E_i) = \sum_{i=1}^{\infty} \mathbb{P}(E_i)$"

Notice,

sum rule ($\because S$: finite, only finitely many $E_i \neq \emptyset$, i.e. we can apply the sum rule)

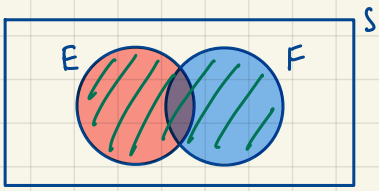
$$\mathbb{P}(\bigcup_{i=1}^{\infty} E_i) = \frac{|\bigcup_{i=1}^{\infty} E_i|}{|S|} = \frac{\sum_{i=1}^{\infty} |E_i|}{|S|} = \sum_{i=1}^{\infty} \frac{|E_i|}{|S|} = \sum_{i=1}^{\infty} \mathbb{P}(E_i) \quad \square$$

# INCLUSION-EXCLUSION PRINCIPLE

## PROPERTY 5

For any (can be not disjoint) two events $E, F \subseteq S$, $\mathbb{P}(E \cup F) = \mathbb{P}(E) + \mathbb{P}(F) - \mathbb{P}(E \cap F)$

Proof




E F $E \cap F$ $E \cup F$

Consider the mutually exclusive events $E \cap F$, $E \setminus F$, $F \setminus E$

We know: $(E \cap F) \cup (E \setminus F) \cup (F \setminus E) = E \cup F$

$(E \cap F) \cup (E \setminus F) = E$

$(E \cap F) \cup (F \setminus E) = F$

By finite additivity, $\mathbb{P}(E \cup F) = \mathbb{P}(E \cap F) + \mathbb{P}(E \setminus F) + \mathbb{P}(F \setminus E)$

$= [\mathbb{P}(E \cap F) + \mathbb{P}(E \setminus F)] + [\mathbb{P}(E \cap F) + \mathbb{P}(F \setminus E)] - \mathbb{P}(E \cap F)$

$= \mathbb{P}(E) + \mathbb{P}(F) - \mathbb{P}(E \cap F) \quad \square$

## COROLLARY

$\forall E, F \subseteq S$, $\mathbb{P}(E \cup F) \leq \mathbb{P}(E) + \mathbb{P}(F)$